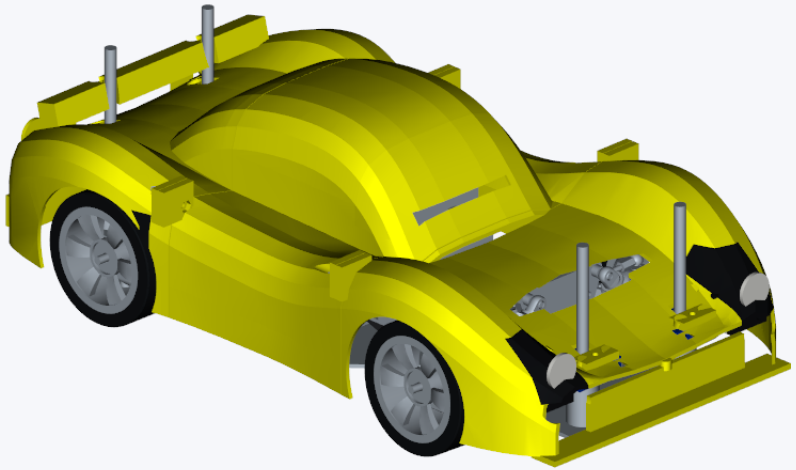
Your Retro GT, in clear steps.

17 printed pieces. Removable white lenses. Four removable post mounts.

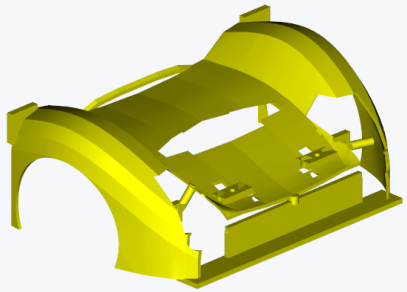


ILLUSTRATED ASSEMBLY GUIDE

Read the fit-test notes before printing. This guide uses the actual modular CAD parts.
127 original mechanical parts remain exact. The unused ProSight FPV antenna is omitted.

Find your parts.

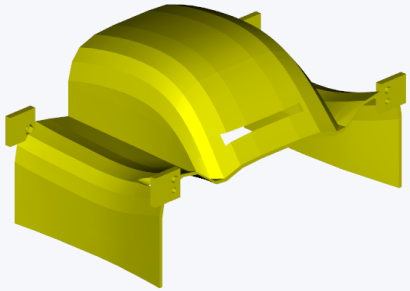
Match the printed labels to the STL filenames in the Retro modular print kit.



B1 / FRONT

127.1 x 200.4 x 75.6 mm

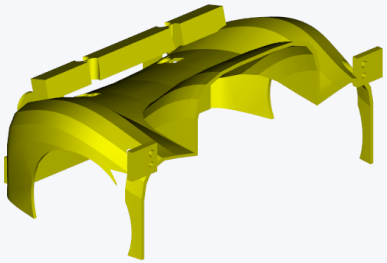
Retro_Front_Module.stl



B2 / MIDDLE

123.5 x 200.0 x 104.8 mm

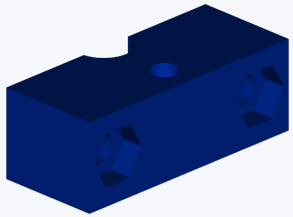
Retro_Middle_Module.stl



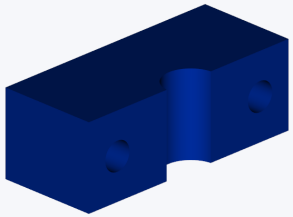
B3 / REAR

128.0 x 200.9 x 98.7 mm

Retro_Rear_Module.stl



O1-O4 / OUTER HALF



I1-I4 / INNER HALF

4 pairs / 8 clamp pieces

Post 1: Retro_Clamp_1_Outer.stl
Retro_Clamp_1_Inner.stl

Post 2: Retro_Clamp_2_Outer.stl
Retro_Clamp_2_Inner.stl

Post 3: Retro_Clamp_3_Outer.stl
Retro_Clamp_3_Inner.stl

Post 4: Retro_Clamp_4_Outer.stl
Retro_Clamp_4_Inner.stl

Find the six lamp pieces.

Two mirrored cartridges: black housings, white lenses and black removable keepers.

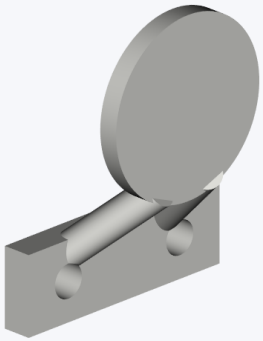


H / BLACK HOUSING

2 pieces: left + right

Retro_Headlamp_Housing_L.stl

Retro_Headlamp_Housing_R.stl

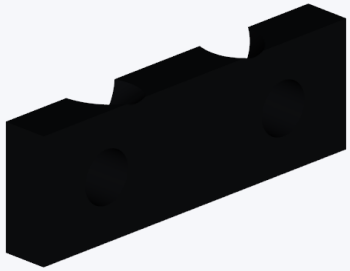


L / WHITE LENS

2 pieces: left + right

Retro_Headlamp_Lens_L.stl

Retro_Headlamp_Lens_R.stl



K / BLACK KEEPER

2 pieces: left + right

Retro_Headlamp_Keeper_L.stl

Retro_Headlamp_Keeper_R.stl

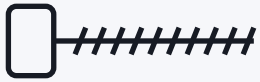
Keep each L set together and each R set together. The pictures show the left-hand set.
The lens flange is captured by its keeper and metal screws. No glue is used.

Gather the hardware.

Metal hardware only. STL files are the PLA parts; do not print the metal references.

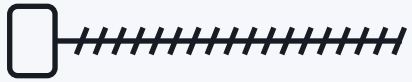
12 x
M3 x 12

4 PANEL SEAMS + 8 LAMPS



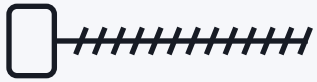
8 x
M3 x 25

CLAMP CLOSURE



4 x
M3 x 16

SHELL SERVICE



24 x M3 hex nuts

Start with 5.5 mm across flats, about 2.4 mm thick. Pockets are 5.8 mm across flats x 2.6 mm deep.

24 x M3 plain washers

About 7 mm outside diameter: 16 x 0.5 mm for seams / clamps / service; 8 x 0.8 mm for lamps. One per screw head.

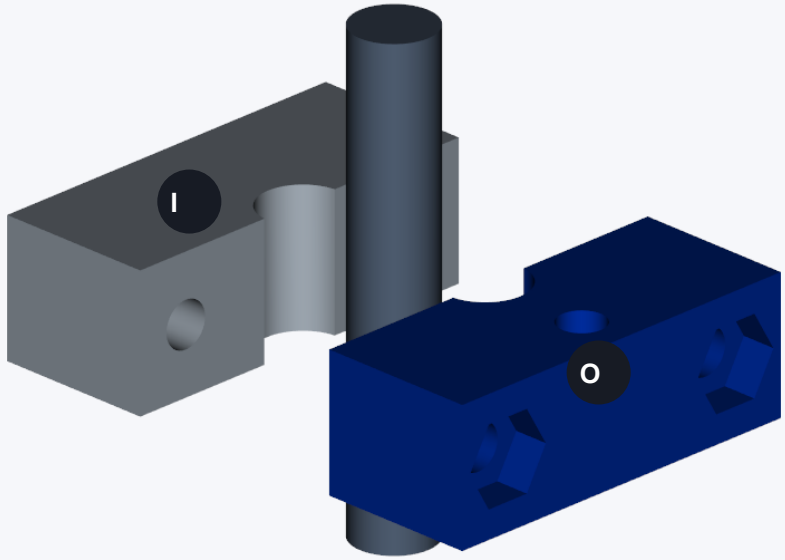
TOOLS

Correct hex driver, calipers, deburring tool and support-removal tools. Keep a small screw-length assortment for the physical fit test.

Screw length is measured below the head. Drawings on this page are not a scale ruler.
Use hand tightening. No safe tightening torque has been established for these PLA parts.

1 Print the fit samples first.

Use the same PLA, nozzle, layer settings and orientation planned for the finished parts.



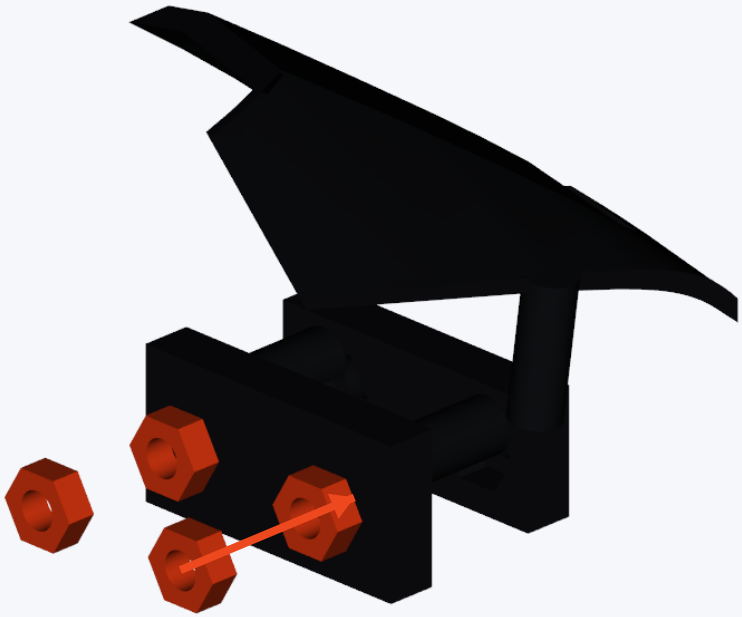
Try the 6.0 / 6.2 / 6.4 mm post bores and 0.2 / 0.3 / 0.4 mm pin clearances; start at 6.2 / 0.3 mm.
A1 nominal bed: 256 mm cubed. CAD envelopes are below 230 mm; orientation, supports and brims need slicer checks.

2 Load the lamp nuts.

Keep the housing separate and its nut pockets accessible. Repeat for L and R.

ADD IN THIS STEP

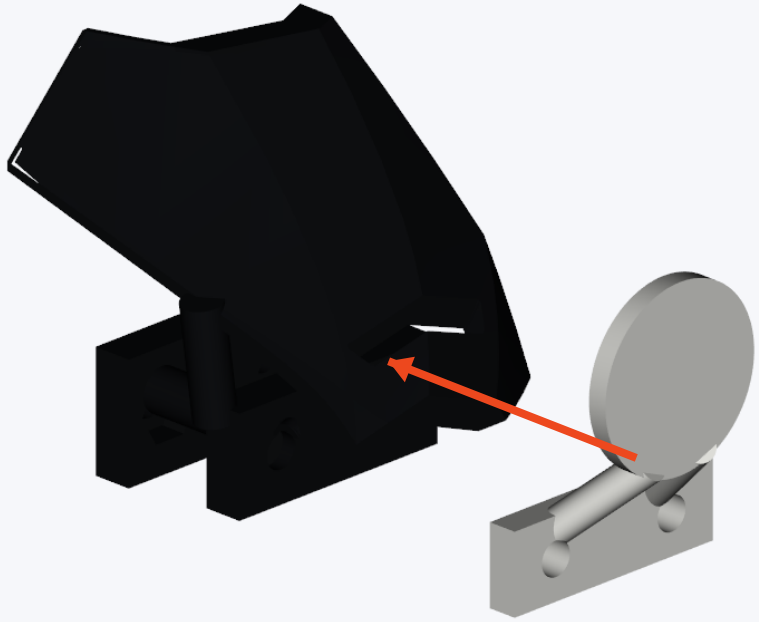
4 nuts per lamp / 8 total



Seat two M3 nuts behind the lens seat and two in the cartridge mounting plate of each black housing. Check all eight nuts sit fully inside the hex pockets before inserting lenses or mounting the cartridges.

3 Insert the white lens.

One lens into its matching black housing. Repeat for L and R.



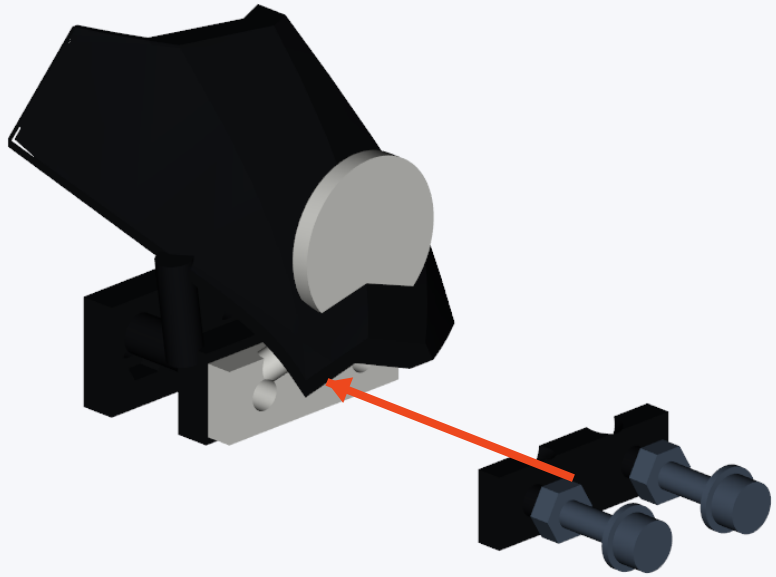
Place the white lens flange against the housing seat. Align the two screw holes and the relieved stalk openings.
Keep the flange flat in the seat. Check the visible white lens sits correctly before adding the keeper.

4 Bolt on the black keeper.

Capture the white flange. Repeat for L and R.

ADD IN THIS STEP

4 x M3 x 12 / 0.8 mm washers



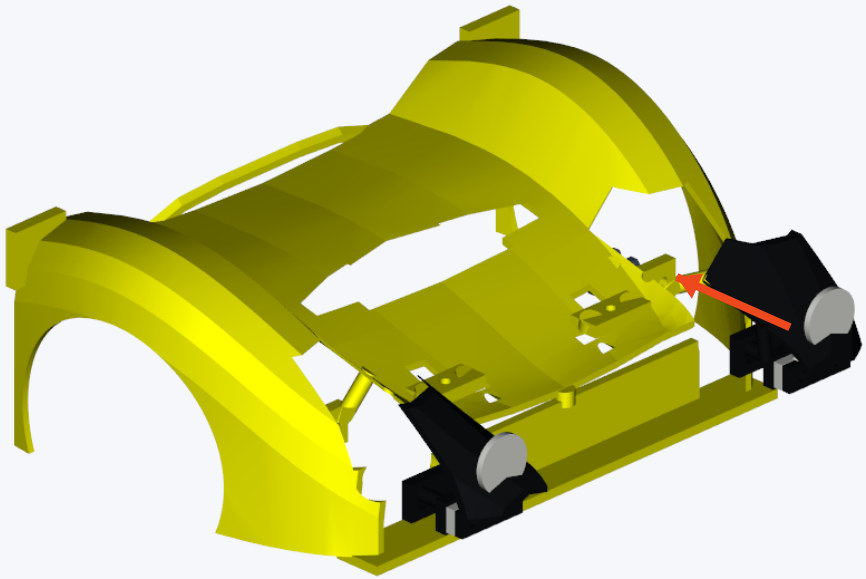
Place the black keeper over the white flange. Fit two screws and 0.8 mm washers into the seated nuts per lamp. Tighten alternately by hand. The keeper must capture the flange without cracking or crushing the lens.

5 Mount both lamp cartridges.

Fit the lamps while the front body module is separate.

ADD IN THIS STEP

4 x M3 x 12 / 0.8 mm washers



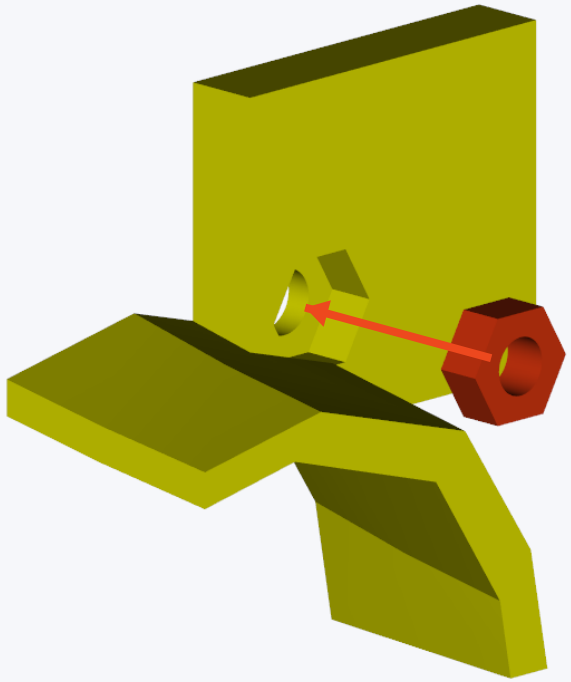
Seat each assembled cartridge on its front-module receiver. Fit two screws and 0.8 mm washers per cartridge. Check head access from inside B1. For lens service, remove the cartridge first, then unbolt its keeper.

6 Seat the panel nuts.

Work from the inside while the panels are separate.

ADD IN THIS STEP

4 x M3 hex nuts



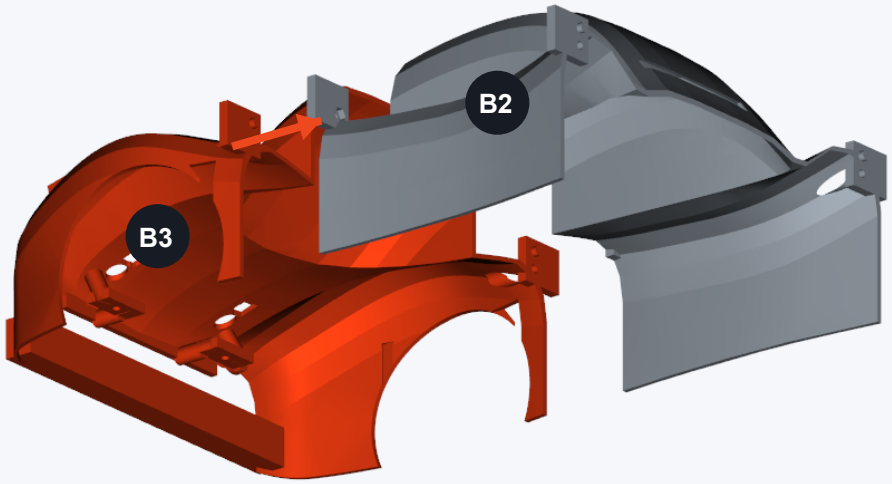
Insert one nut into each of the four seam pockets: two at the rear/middle seam and two at the middle/front seam.
The nut must sit fully inside its hex pocket. Do not force a tight nut or split the pocket wall.

7 Join rear to middle.

Underside view. Orange highlights the part being added.

ADD IN THIS STEP

2 x M3 x 12 + 2 washers



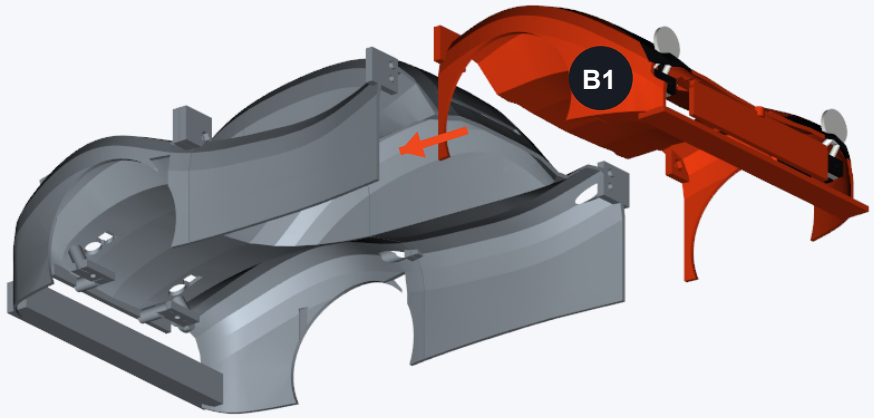
Locate the two pins in their matching sockets. Fit one screw and washer per side into the captive nuts. Keep both screws loose until the outside seam aligns, then tighten progressively by hand.

8 Add the front module.

Keep underside access to the screw heads.

ADD IN THIS STEP

2 x M3 x 12 + 2 washers



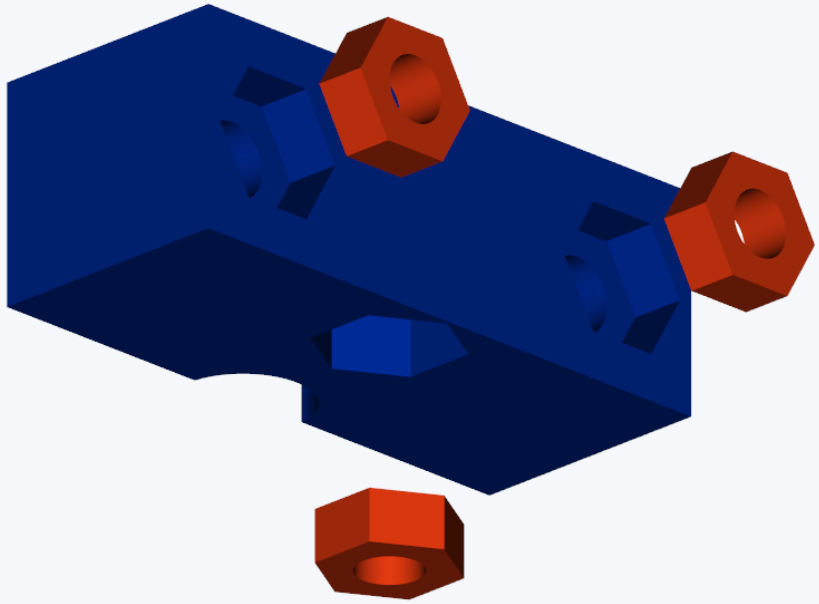
Locate B1 on B2, then fit the remaining two seam screws and washers.
Check the four joints for gaps, cracking and accessible screw heads before putting the shell on the car.

9 Load the clamp nuts.

Repeat this operation for O1, O2, O3 and O4.

ADD IN THIS STEP

3 nuts per outer half / 12 total



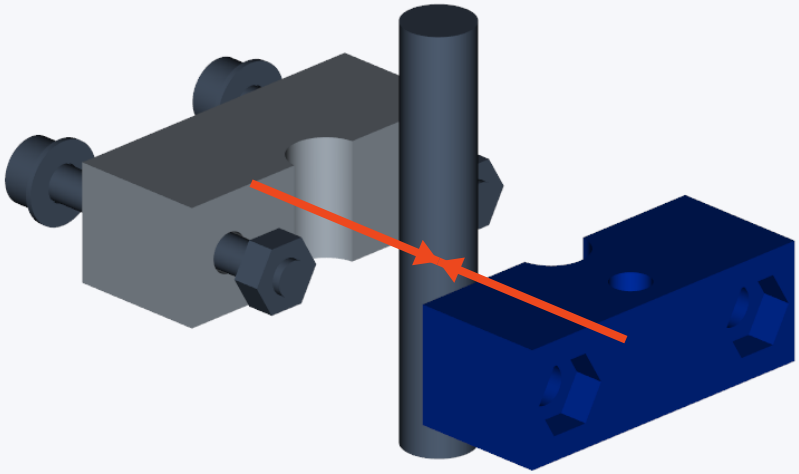
Each outer half receives two nuts in its side pockets and one nut in the underside service-screw pocket. Make sure all three nuts sit flat before the collar goes around a post. The inner half has no nut pockets.

10 Close a collar around a post.

Repeat for four posts. Gray shaft = unchanged original 6 mm post.

ADD IN THIS STEP

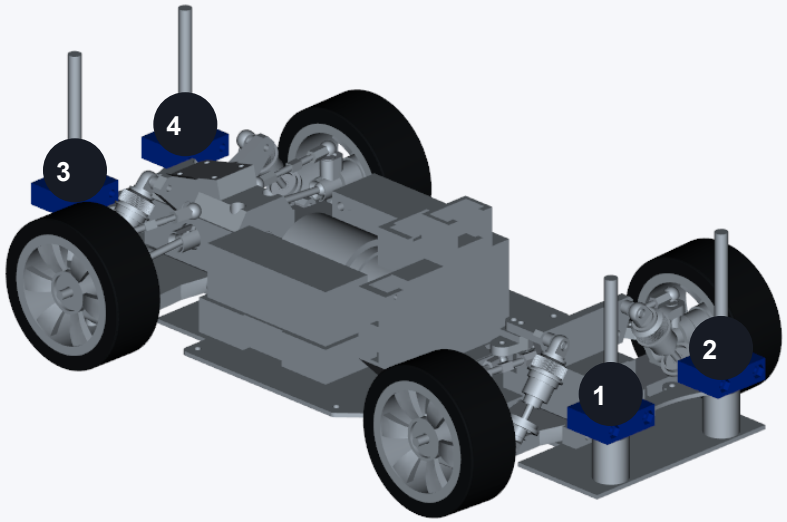
2 x M3 x 25 + 2 washers



Bring the rigid halves together around the post. Fit the two transverse screws and washers into the captive nuts. Tighten evenly only until the collar stops slipping. Keep the split gap; do not crush the PLA to close it.

11 Position all four collars.

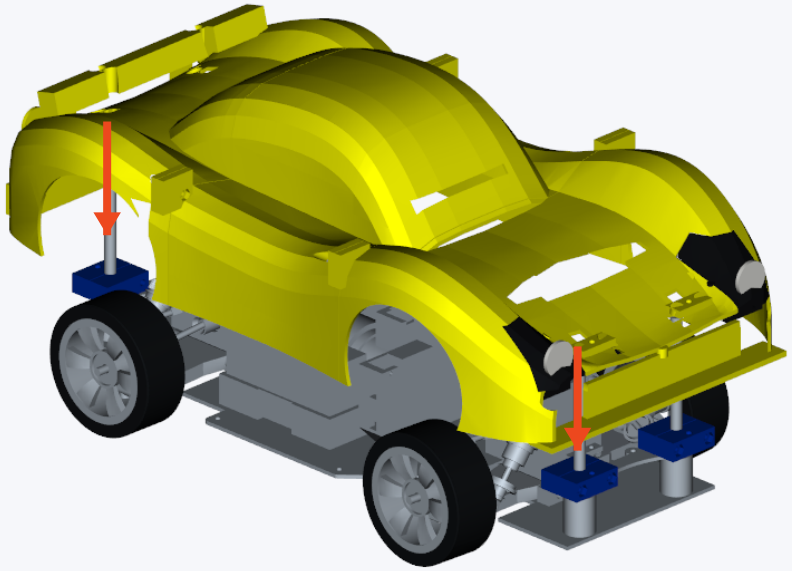
Front posts: X=159. Rear posts: X about -159. Lateral position: Y=+/-32.75 mm.



Set each collar between Z=25 and 35 mm in the model frame. The collar bottom sits 1.2 mm above the original fixing plate.
The half with the vertical service nut faces the end of the car. Confirm all four collars grip without twisting.

12 Lower the shell onto its seats.

The shell is already joined. Keep all wires and original mechanical parts clear.



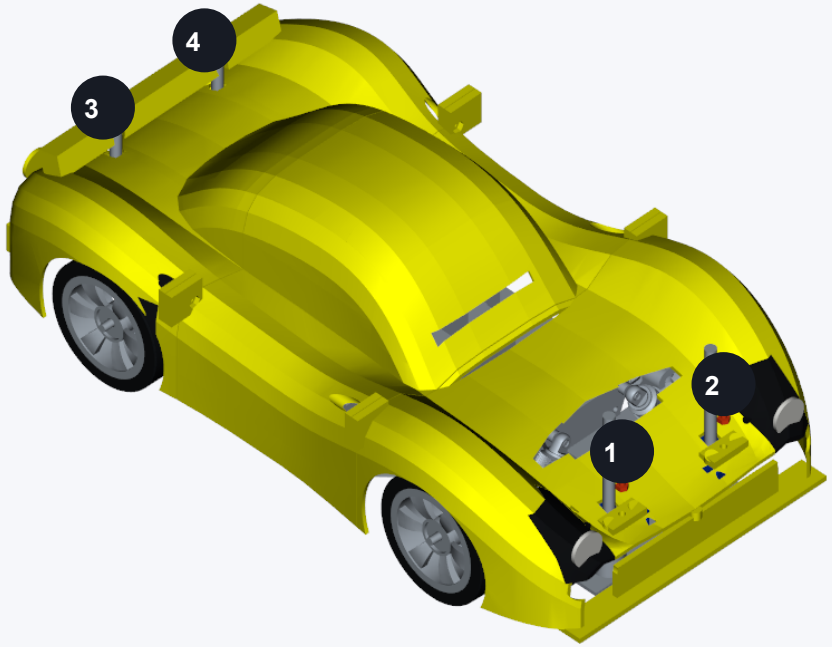
Align the four service-screw openings with the outer collar halves. The modeled seat-to-collar gap is 0.3 mm.
If a seat does not align or the shell touches original hardware, stop and adjust the fit. Do not pull it down with a screw.

13 Fit the four service screws.

Insert from above through the access openings.

ADD IN THIS STEP

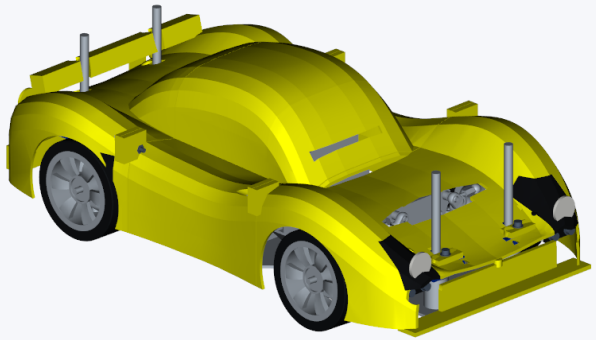
4 x M3 x 16 + 4 washers



Place one washer under each screw head and engage the nut in the collar below. Tighten evenly by hand.
For service removal, undo these four screws and lift the joined shell away. The collars can stay on the posts.

14 Check before driving.

A CAD fit check does not establish printed strength, clamp grip or crash performance.



1 Check full left / right steering and wheel clearance.

2 Check full suspension compression and droop.

3 Check lamp retention, wiring and screw-tip clearance.

4 Hand-test each seam and collar for movement.

5 Run slowly; inspect for loose screws or cracks.

6 Recheck after use. Keep PLA away from high heat.

M3 is a reasonable prototype size; PLA roots, nut walls and collar friction may fail before the steel screws.
This is a fit-test guide, not a customer-release load rating. Revise the design if the physical checks fail.